

Sumit Mishra

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Research Focus

Deep learning researcher specializing in **multimodal foundation models** and **vision-language learning** for safety-critical applications. Extensive experience with CLIP-based systems, cross-modal knowledge transfer, and explainable AI. Strong publication record at top-tier venues (IEEE Transactions) bridging foundational research with real-world deployment.

Education

Ph.D. in Robotics 2023–2026 (Expected)
Korea Advanced Institute of Science and Technology (KAIST), GPA: 3.85/4.3 or 95.5/100 % • *Thesis: Explainable AI for Safer Roads Using Multimodal Framework*

M.S. in Robotics 2021–2023
Korea Advanced Institute of Science and Technology (KAIST), GPA: 4.0/4.3 or 97/100 % • *Thesis: Multi-modal Data Usage for Smart City Mobility Services*

B.Tech. in Electronics and Communication 2012–2016
Galgotias College of Engineering and Technology (AKTU)

Core Technical Expertise

Vision-Language Foundation Models: CLIP, DINO, Florence, BLIP, LLaVA | Concept bottleneck models | Zero-shot/few-shot learning | Prompt engineering | Cross-modal alignment

Multimodal Learning: Cross-modal knowledge distillation (thermal→optical, vision→text) | Contrastive learning (SimCLR, MoCo, text-anchor-based) | Weak Supervised learning | Multi-instance learning under weak supervision

Deep Learning & Computer Vision: PyTorch (Primary), TensorFlow | Transformers | Vision Transformers (ViT, DeiT) | Attention mechanisms | Dilated temporal convolutions | OpenCV, Detectron2, MMDetection | CAM/Grad-CAM for explainability | Self-supervised learning | Domain adaptation | Model compression | Edge-AI optimization

Programming & Tools: Python (Expert - 8+ years) | CLIP, Hugging Face | Git, Docker, Weights & Biases, TensorBoard, Jupyter

Work Experience

Remote Research Intern Jun 2025–Nov 2025
Supervised by Dr. Pranjay Shyam, Applied Scientist II, Amazon Fulfillment Technologies & Robotics, Email: pranjay@amazon.com
Built zero-shot anomaly detection (92.42% accuracy) with Triple-CLS architecture for cross-modal alignment; Evaluated foundation models (CLIP, BLIP, LLaVA) for edge deployment analyzing accuracy-efficiency trade-offs

Graduate Research Assistant Mar 2021–Present
Vehicular Intelligence Lab, KAIST, Advisor: Prof. Dongsoo Har
Leading 7 papers on multimodal learning (2 under review: IEEE T-ITS, IEEE T-ETCI; 5 published in IEEE journals IF: 4.3-9.1); Developed 3 CLIP-based frameworks: VisText-Net (CBMs), TIME-VAD (text-informed learning), wildfire detection (cross-modal KD); Achieved SOTA (95-100% accuracy) across 6 datasets with edge-deployment feasibility (2-3 GFLOPS)

Research Consultant Jun 2018–Feb 2021
Learnogether Technologies Pvt. Ltd.
Led smart city AI projects focusing on traffic management and urban mobility; Collaborated with Deloitte India on traffic systems; Secured £11,500 grant from University of Edinburgh for MobilitySafe project; Received 12.5 lakh INR grant from MSME, India

Research Scholar Aug 2016 - Jun 2018
Galgotias College of Engineering and Technology
Led development of real-time congestion mapping methodology similar to [Google project](#) for Delhi region. Conducted POC for hardware/software prototype development for traffic light control board, IoT network security, and low-cost visibility measurement technology.

Key Research Project and Related Papers

Self-Supervised Visual Language Learning for Explainable Accident-Prone Feature Discovery

Mishra, S., Mishra, M., Ko, KW., and Har, D. [Under Review - IEEE T-ITS]

- Novel dual-branch CLIP-based concept bottleneck model (CNN + CLIP text encoder); Automated concept discovery via mechanistic interpretability of linear perceptron weights; counterfactual validation (**64.25% confidence drop** when modifying top 5% concepts)
- Optimal transport with Sinkhorn normalization for patch-to-concept assignment; Class Activation Map based ablation analysis showing **39-63% confidence reduction** when masking high-attention regions

TIME-VAD: Text-Informed Magnitude Enhancement for Vehicle Accident Detection

Mishra, S., Mishra, M., Shyam, P., and Har, D. [IEEE Transactions on ITS, IF: 9.1]. Featured in [KAIST Breakthrough](#).

- Reversed CLIP magnitude assumption: accident frames show lower magnitudes (15.38) vs normal (17.93); Used Magnitude-enhanced Multiple Instance Learning using least-K features under weak supervision; Used Text-anchor based contrastive loss; Results: 94.44% ROC-AUC on DoTA (9.74% over SOTA), 99.97% AP, 4.99s mTTA on CCD
- Proposed Dilated Temporal Conv-Attention (DTCA) block: pyramid dilated convolutions (rates: 1,2,4) + 3-path attention; Cross-domain generalization: CCD→DAD 95.50% (0.12% over in-domain); Multi-dataset performance: CCD 95.38% vs 91-94% baseline, DAD 84.43% vs 75%, DoTA 94.44% vs 84.7%

Zero-Shot Road Anomaly Detection Based on Text Corpus Knowledge and Segmentation-First Localization

Mishra, S., Kumar, H., Shyam, P., Har, D., and Ko, K.W. [Under Review - IEEE Transactions on ETCI]

- Triple-CLS architecture: controlled noise injection for cross-modal synchronization (vision-language); Zero-shot captioning without paired training data; Performance: 92.42% accuracy, 16-22% over VLMs (LLaVA, BLIP), 15-25× smaller model size; Efficient pipeline: semantic segmentation → CLIP classification
- Systematic foundation model evaluation: CLIP variants, BLIP, LLaVA; Trade-off analysis: accuracy-latency-size for edge deployment; Prompt engineering strategies for safety-critical zero-shot Notification Generation using GPT2 and LLM as judge evaluation

Cross-Modal Distillation for Real-time Wildfire Detection and Localization in Edge-Deployed Aerial Vehicles

Mishra, M., Mishra, S., and Shin, H. [ISPRS Journal of Photogrammetry and Remote Sensing, IF: 12.9]

- Thermal-to-optical knowledge transfer: CLIP teacher (frozen) → CNN student (trainable); Spatial feature distillation via Pearson correlation (ρ_{hw}) for structural pattern transfer; Semantic-features relation distillation via cosine similarity affinity matrices; Edge-AI optimized: 2.99 GFLOPS, 0.004s inference, 90.96% patch-level, 100% image-level accuracy
- Used Learnable text-anchor contrastive loss; Focal loss for class imbalance along with Multi-loss optimization; Performance: FNR: 14.78%, FPR: 8.82%

Additional Selected Publications

[IEEE Transactions on ITS 2023, IF: 9.1] *Mishra, S., Rajendran, P K., Santos V, L F., and Har, D.* – “Sensing accident-prone features in urban scenes for proactive driving and accident prevention”. Proactive ADAS: CLIP+DenseNet (92% acc). POC deployed with Deloitte India (Lucknow). Featured in [KAIST Breakthrough](#).

[IEEE SPL 2026, IF: 3.9] *Mishra, S., Kumar, H., Ko, K.W., and Har, D.* – “IMUDistill: Knowledge Transfer for Enhancing Low Precision IMU Performance”.

[IEEE ITS Magazine 2025, IF: 5.5] *Mishra, S., Sengar, N., and Har, D.* – “A Secure, Blockchain-Enabled Vehicular Sensor Communication Protocol With Deep Learning-Assisted Anomaly Detection”.

[IEEE Sensors Journal 2024, IF: 4.3] *Mishra, M., Mishra, S., and Har, D.* – “Integrating Multi-sourced Sensor Data for Enhanced Traffic State Estimation”.

[IEEE IV 2022] *Shyam, P., Mishra, S., Yoon, K J., and Kim, K S.* – “Infra Sim-to-Real: An efficient baseline and dataset for Infrastructure based Online Object Detection and Tracking using Domain Adaptation”.

[ECCV 2022] *Rajendran, P K., Mishra, S., Santos V, L F., and Har, D.* – “RELMOBNET: A Robust Two-Stage End-To-End Training Approach for MOBILENETV3 Based Relative Camera Pose Estimation”.

[arXiv] *Mishra, S., Mishra, M., Kim, T., and Har, D.* – “RoadSafe: A Road Redesign Technique Achieving Enhanced Road Safety by Diffusion Model Inpainting” (arXiv:2302.07440).

Patents & Deployments

Indian Patent No.: 538566 (Granted 2024) – Gupta, A. and **Mishra, S.** “System for spatio-temporal traffic data reporting for smart city services”. Implemented in Hyderabad & Bangalore, India for adaptive traffic signaling.

Korean Patent Applications (Pending, KAIST-supported filing):

- (1) **Mishra, S., Mishra, M., Har, D.** “Real-time traffic monitoring and analysis system” (10-2024-0103095)
- (2) **Mishra, S., Mishra, M., Har, D.** “Accident probability prediction system” (10-2024-0087193)

Deployed Systems: Multimodal navigation portal for COVID-19 safe routing (Deloitte, GIZ, University of Edinburgh, £11,500 grant); Accident-prone feature notification system POC (Deloitte India, Lucknow region)

Awards & Grants

- Full-funding KAIST scholarship for MS & PhD (2021–Present) • £11,500 grant from University of Edinburgh for MobilitySafe project (2020) • 12.5 lakh INR project grant from MSME, Government of India (2018)

Academic Service

Conference Reviewer: IEEE Intelligent Vehicles Symposium (IV), ACM CHI

Journal Reviewer: IEEE Transactions on Intelligent Transportation Systems, IEEE Sensors Journal, IEEE Access, Mobile Networks & Applications, Multimedia Tools & Applications, Scientific Reports, Cluster Computing

References

Prof. Dongsoo Har (PhD Advisor) – Professor, The CCS Graduate School of Mobility, KAIST, dshar@kaist.ac.kr, [Website](#)

Dr. Pranjay Shyam (Research Collaborator) – Applied Scientist II, Amazon Fulfillment Technologies & Robotics, Seattle, USA, pranjay@amazon.com, [LinkedIn](#)

Dr. Luiz Felipe Vecchietti (Co-author) – Senior Researcher, Max Planck Institute for Security and Privacy (MPI-SP), Germany, felipe.vecchietti@mpi-sp.org, [Website](#)

Prof. Ankit Gupta (Patent Co-inventor) – Professor and MoRTH Chair Professor, Transport Engineering Section, Department of Civil Engineering, IIT (BHU), ankit.civ@iitbhu.ac.in, [Website](#)